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Abstract

This research focuses on the issues associated with hot-hole blasting in South African surface coal mines including safety, compliance with laws, explosives choice, blast design, operation expenses and recommendations for hot holes management. Compliance with MHSC standards was assessed in order to enable safe blasting in reactive rock formations. Blast design using 165 mm drill hole and emulsion explosives resulted in good fragmentation and vibration data.

Inhibited explosives were found to be the safest explosives having the highest thermal stability. The total annual blasting cost was calculated as R49.1 million. It is concluded that proper monitoring, explosives choice and blast design are required for safe and efficient hot hole blasting operations.

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1. Introduction

The mining industry in South Africa faces several problems, from geological problems to dangerous conditions such as "reactive" or "hot" ground, which are frequently experienced in the Witbank and Mpumalanga coal mines as well as in the Northern Cape. If this situation is not handled correctly, it could lead to toxic fumes being produced, early detonations, and uncontrollable fires. It could also cause poor fragmentation of the blast, delays in production, and high costs of blasting (AEISG, 2020; Tose, 2022). Hot holes also pose a significant health risk to miners who may be exposed to toxic gases such as carbon monoxide, sulphur dioxide, and carbon dioxide.

While temperature monitoring equipment and special methods of blasting are used in mines to address these challenges (Anthony and Grobbelaar, 2009), hot holes still represent a significant problem because of unexpected variations in the temperature of blast holes. As a result, new methods must be developed that can monitor and predict hot holes in real time. Consequently, this report will explore the issue of hot holes in surface coal mines. The factors behind the formation of hot holes, regulations regarding explosives in South Africa, the choice and use of explosives, the blasts design in hot ground conditions and assesses methods of monitoring to ensure safe and efficient mining operations in surface coal mines in South Africa.

2. Literature review

2.1 Formation Mechanism

What are hot holes?

According to Mpofu et al. (2024) hot holes are short holes that have been drilled, which have an in hole ambient temperatures of 40 degrees Celsius or above or show a temperature increase of 3 degrees or more during the monitoring process.

Physical processes leading to the formation of “hot holes”

The processes involved in the formation of hot holes are mainly linked to the presence of old underground mine workings under active surface coal mines. As reported by Mpofu et al. (2024), cracks, cavities, sinkholes, and blast holes provide passageways through which oxygen gets access to the underground workings and heat/gases escape towards the surface. The underground openings were formed as a result of earlier mining operations whereby low seams were mined underground prior to surface mining of upper seams. As a result of being enclosed, the heat produced inside the underground workings is trapped inside the rock mass thereby heating up gradually with time. When drilling operations encounter these heated areas, hot holes with excessively high temperatures are created. However, the situation worsens when drilling takes place in the coal seam as coal provides an increased area for heat production and dissipation.

Chemical processes leading to the formation of “Hot holes”

The reactions that lead to the creation of hot holes are mainly related to spontaneous combustion and oxidation reactions. Oxygen from the external environment entering the workings via fractures and blast holes causes a reaction with coal surfaces. The oxidation process is an exothermic chemical reaction, where heat energy is emitted in the process. The oxidation process is accompanied by the build-up of heat in the workings since it is difficult to dissipate the heat. As a result, the emission of heat leads to an increase in the temperature of coal and rock mass, thus, contributing to the occurrence of underground fires and hot ground. According to Mpofu et al. (2024), the rate of oxidation increases with

the increase of the coal surface area coming into contact with oxygen, thus, explaining the frequent occurrence of hot holes in coal seams rather than overburden. Furthermore, reactive ground with iron and copper sulphides can cause a reaction with explosive material, like ammonium nitrate. Chemical reactions between sulphide mineral and nitrate compounds also emit heat. (Mpofu et al., 2024).

2.2 Legislative Framework (South Africa)

In relations to the explosives and fire/explosion prevention, the DMRE and MHSC guide on the Prevention of Flammable Gas Explosions in Mines, require mines to classify the areas that pose risks, monitor flammable gas levels and take preventive measures from fires and explosions (MHSC, 2021). These requirements are relevant to hot hole management in mining as underground heating might lead to the occurrence of flammable gas ignition while performing drilling and blasting operations (MHSC, 2021). The best practise guideline for hot hole management requires the identification of the classification of mining blocks based on the potential of containing hot holes before undertaking any drilling and blasting work. Once the drilling is done, temperatures in the blast holes should be measured, recorded and categorised depending on their temperatures while those with smoke, fumes and excessive heat should be labelled and acted upon accordingly (MHSC, 2021).

In addition, the regulation requires that the blast hole temperatures be rechecked prior to charging in order to confirm that the holes meet the correct temperature class. It also requires the application of explosives for hot holes only, in addition to monitoring continuously throughout the charging and blasting process. Any hot hole considered unsafe for charging should be sealed up using the mine's risk assessment protocol in order to prevent any chance of an explosion or fire outbreak (MHSC, 2021).

2.3 Explosive Selection

What is a reactive ground?

Reactive ground refers to geological material containing elevated sulphide minerals which can react exothermically when exposed to ammonium nitrate-based explosives (SAFEX International, 2025). This reaction causes rapid heat build-up, toxic gas release, and may result in premature detonation.

Standard explosives used in mining

The standard explosives commonly used in mining are ANFO and emulsion explosives.
ANFO is composed of porous ammonium nitrate prills mixed with fuel oil and is primarily used in dry blast holes because of its water resistance. According to Djerdjev et al. (2021) in reactive ground conditions, ANFO presents significant safety concerns because sulphide minerals can accelerate their decomposition, leading to self-heating and an increased risk of unintended detonation. In contrast, emulsion explosives are water resistant explosives made of oxidiser droplets dispersed in a fuel phase. These explosives are stable and effective in wet/dry conditions (Pearton, 2015). However, in reactive ground, the NH_4NO_3 component within the emulsion explosion may still react with sulphide minerals, increasing heat generation, gas release and high risk of premature initiation if inhibitors are absent (Safex, 2025).

Inhibited explosives

To mitigate these risks, inhibited explosives are used. These are specialised blasting agents designed for reactive or high temperature ground conditions (BME, 2022). They contain additives such as urea, which neutralise acids, slow exothermic reactions, and prevent runaway NH_4NO_3 decomposition. In contrast to standard explosives, inhibited explosives offer improved thermal stability, reduced chemical reactivity, and a significantly lower risk of premature detonation in reactive environments.

Table 2: Comparison between Standard Explosives and Specialised Explosives (Djerdjev et al., 2021 and BME, 2022)

Criteria	Standard explosives (ANFO/Emulsions)	Specialised explosives (BME's INNOVEX RG)
Reactive ground behaviour	Prone to chemical reactions with sulphides	Reduced reactivity
Heat resistance	Becomes unstable in hot holes	Designed for higher thermal stability
Premature detonation risk	Higher in reactive environments	Significantly reduced
Application	General mining conditions	Reactive/hot ground conditions

2.4 Hot Hole Blasting Protocols.

Temperature Measuring Techniques

Hot hole blasting requires strict controls to prevent explosive degradation, premature initiation, and safety hazards (Mpofu, 2022). Explosives are designed to operate within specific temperature ranges. Each explosive product has a manufacturer defined maximum loading temperature indicating the highest safe operating limit. When these temperature limits are exceeded, explosives such as ammonium nitrate based may undergo thermal decomposition, increasing heat and the risk of premature detonation (Bourne, 2022). Consequently, temperature monitoring is essential in hot hole blasting operations. Temperature measurements are commonly conducted using monitoring devices such as thermocouple probes as shown in figure 4.1 and handheld infrared thermometers shown in figure 4.2 to ensure that blast hole conditions remain within safe operational limits prior to explosive loading.

Sleep-time restrictions

Sleep time refers to the period explosives remain loaded in hot blast holes before detonation, and it should be kept as short as possible to reduce the risk of explosive degradation, premature initiation, and hazardous chemical reactions in reactive or high temperature conditions (Mpofu et al., 2024). At existing reactive ground sites, the recommended maximum sleep time is generally limited to one week, based on laboratory testing assessing explosive stability under reactive conditions. If explosives must remain in blast holes beyond this period, a comprehensive risk assessment must be undertaken collaboratively by the explosive supplier and the mining operation (AEISG, 2017).

Specialised stemming requirements

According to Oates and Spiteri (2021), stemming is important in blasting because it confines explosive gases within the blast hole, directing more energy into rock breakage instead of allowing it to escape through the collar. In reactive ground conditions, the stemming material used should be inert to prevent chemical reactions within the explosive environment. The timing of stemming is also critical, as delays increase the exposure time of explosives to reactive conditions and raise the likelihood of adverse reactions. Prolonged stemming operations can increase explosive sleep time and compromise operational safety therefore, stemming should be completed efficiently using suitable inert materials.

3. BLAST DESIGN

The aim of this blast design is to achieve adequate rock fragmentation while minimising ground vibrations due to the close proximity of nearby buildings and infrastructure. The rock mass to be fragmented consists of soft sandstone. The initial blast design procedure applied in this study follows the methodology provided in the study guide. A drill hole standard deviation of 0.3 m was assumed and used throughout the calculations. The assumption is based on the fact that high accuracy equipment is used and minimal drill hole deviation occurs. The table below presents the data utilised in the initial blast design calculations.

Bench height (BH)	12m
Technical powder factor of sandstone	0.3Kg/m^3
Burden Ratio (a)(Micromine,2025)	1.15
Rock Factor (A)	9
Drill hole diameter (D)	165mm
Block length	200m
Block Width	50m
Density of emulsion explosive (ρ)	1.2g/m^3
Standard deviation of Drill holes(W)	0.2m
Distance to closest buildings	600m

The ideal blast pattern for the fragmentation of sandstone is a staggered pattern therefore, a burden ratio of 1.15 is adopted. This burden ratio promotes efficient rock breakage whilst producing a relatively uniform fragmentation. The selected fragmentation size is appropriate because the primary objective of the blast is to expose the underlying coal seam (Micromine,2025).

3.1 Calculations of the dimensions of the blast design

Mass per Meter

$M = \rho D^2/1273$

= 1.2x165^2/1273

= 25.66

Steaming length

SL = D x 30

= 165 x 30

= 4950mm

=4.95m

Burden

$$B = \sqrt{\left[\frac{M \times (BH - SL)}{a \times BH \times PFtech}\right]}$$

$$= \sqrt{\left[\frac{25.663 \times (12 - 4.95)}{1.15 \times 12 \times 0.3}\right]}$$

= 6.61m

Spacing = 1.15 x B

= 7.60m

Sub drill

SD = 1/4xB

= 1/4 x 6.61

= 1.65m

By virtue of the nature the rock being soft rock, the subdrill calculated above is taken to be negative.

Determination of number of holes

Number of holes along the length = 200/6.61 = 30.26

= 30 holes

Number of holes along width = 50/7.60 = 6.58

= 7 holes

Total number of holes = 7 x 30= 210 holes

Mean Fragmentation size

The mean fragmentation size is determined by the Kruz Ram method.

Q = 25.66x (12-4.95-1.65) =138.56

$X_{50} = A(K)^{-0.8}Q^{0.167}(115/E)^{19/20}$

$= 9(0.3)^{-0.8}(138.56)^{0.167}(115/116)^{19/20}$

= 53.29cm

Uniformity index

This indicates the uniformity of the fragment size.

Based of quantitative analysis of drill hole data, an estimated value of 0.3m of the standard deviation has been taken. L=BH-SL-SD

L= 12-4.95-1.65 = 5.4 n = (2.2-14(B/D)) x (1-

W/B) x (1+(a-1)/2) x L/H

= (2.2-14(7.60/165) (1-0.3/7.60) (1+0.15/2) (5.28/12)

≈ 0.78

N= 0.78 + 0.078

=0.858

According to the project brief the needed fragmentation is 80%. This can be predicted by the rosin Rammler equation.

$R = 1 – e^{(-\ln 2(x/x_{50})^n)}$

$0.8 = 1 – e^{(-\ln 2)(x/53.29)^{0.858}}$

$-0.2 = – e^{(-\ln 2)(x/53.29)^{0.858}}$

X_{80}= 157.20cm

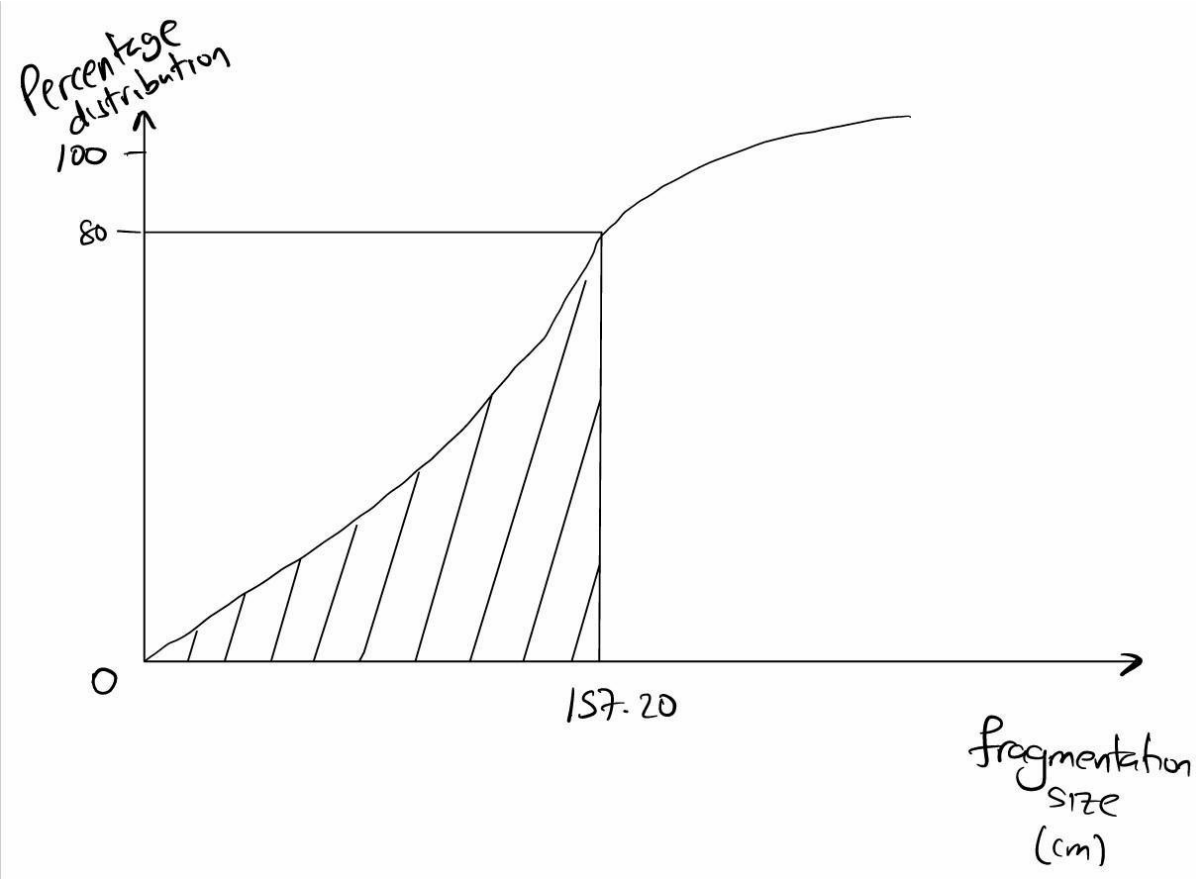


Figure 3.1 A graph of fragmentation distribution for the sandstone blasted.

Initiation Timings

The initiation systems used are electronic initiation systems to ensure precise and accurate times. The programmable detonators are supplied by IMEX. The initiation timings below have been calculated based on the information shown on the lecture slides (MINN2012A).

Intra-row timing: 6ms/m *6.61m=39.99ms=40ms

Inter-row timing: 30ms/m*6.61m= 198.3ms = 198ms

Vibration calculation

The vibration calculation is crucial for the safety of the surrounding buildings. In order to get the PPV, we have used the blast design and mass per hole to determine the number of holes going off at the same time (MINN2012A).

Number of holes going off at the same time = 4 Mass of

explosive per hole = 25.66 x 4.69 = 120.3454

$$PPV = 700[\frac{}{\sqrt{M}}]^{D}{}_{-1.67}$$

= 700[600/(481.3816)^.5]^-1.67

= 2.79m/s

The peak particle velocity obtained is well below the threshold, hence the operation is considered to be safe.

3.2 FINANCIAL ANALYSIS

Budget

Below lies the calculation used to estimate the annual budget for this mining operation.

1. Number of detonators used = number of detonators per blast x number of blast per annum

   = 210 x 96

   = 20160 detonators/annum

   Cost of detonators = R150

2. Cost of explosives per tonne (Coherent market insights,2025) ≈$1000.00 = R16640.90

   assumption = bench dimensions remain the same throughout all the explosives' procedures.

   The actual powder factor is used to determine annual consumption of the explosives. (MINN2012A, page 146)

   PF_{act} = (M x (BH - SL - SD)) / (BH x L x W))

   = (25.66 x (12 – 1.65 – 4.95)) / (12 x 6.61x 7.60)

   = 0.23

Mass of explosive per blast = 0.23 x (200 x 12 x 50) = 27600kg

Number of blasts per year = 96

Annual mass of explosives used = 96 x 27600 = 2592000kg =2649.6t

MATERIAL	UNIT COST	TOTAL NUMBER OF UNITS	SUBCOST
Emulsion explosive	R16640.90/t	2649.6t	R44091728.64
Detonators	R150	20160	R3024000.00
TOTAL COST PER ANNUM			R49089052.80

4. **Recommendations**

Based on the findings in this report:

- It is recommended that the mine utilise inhibited explosives instead of standard explosives, as inhibited formulations provide greater stability under high temperature ground conditions and significantly reduce the likelihood of premature detonation.
- It is proposed that appropriate explosive sleep time in hot-hole conditions be maintained, and that the interval between charging and initiation phases be minimised. This is intended to reduce the likelihood of premature initiation and to mitigate potential degradation or damage to explosive materials under elevated in hole temperature conditions.
- Given the inherently empirical nature of blast design, it is further recommended that burden parameters be adjusted in operational requirements in order to achieve the desired fragmentation outcomes. Such adjustments should be undertaken to control fragmentation size to meet mine production needs.

5. Conclusion

This report explored the issue of hot-hole management in South African surface coal mines by discussing the reasons for forming hot holes, legislation, selection of explosives, blast design, and costs involved. It has been found out that hot holes are primarily created owing to the processes of oxidation and spontaneous combustion, which occur within old underground workings and reactive and high temperature ground conditions. The importance of adhering to the guidelines and procedures set by MSHC have been emphasised to ensure safety from premature detonation, toxic gas emissions, and underground fires. In addition, temperature control, sleep time control, and use of inert stemming material have been pointed out as other safety measures in hot hole blasting.

In regard to the explosives used in hot hole blasting, it has been found that inhibited explosives are better suited to the task because of their superior thermal stability and low chemical reactivity in comparison with ANFO and emulsion explosives. The blast design performed for a soft sandstone bench at the depth of 165 mm using emulsion explosives yielded satisfactory results regarding fragmentation and vibration, with X_{80} fragmentation size being equal to 157.20 cm and peak particle velocity of 2.21 mm/s. The estimated cost per annum for explosives and initiation systems was determined to be R49.1 million. Overall, the study shows that monitoring, proper choice of explosives, adherence to blasting procedures, and blast optimisation are essential for hot-hole blasting to be conducted safely, efficiently, and economically.

Appendix

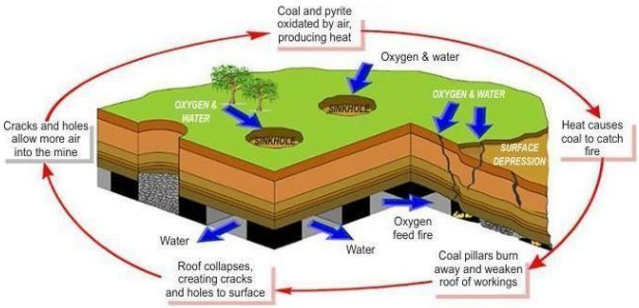


Figure 2: Illustration of the sequence of processes involved in mining above old underground workings (Tose, 2022).



Figure 3.1: UNI-T handheld infrared thermometer (CIB Partners, 2020)



Figure 3.2: PTM-806 Model thermocouple (Griesel, 2021)

References

Australian Explosives Industry and Safety Group (AEISG), 2017. Code of Practice: Elevated Temperature and Reactive Ground. 4th ed. Australia: AEISG. Available at: https://www.saimm.co.za/Journal/v121n08p415.pdf [Accessed 10 May 2026].

BME. (2022). Reactive ground blast management in a South African open-pit zinc mine. [Online] Available at: https://bme.co.za/reactive-ground-blast-management-in-asouthafricaopen-pit-zinc-mine/ [Accessed 11 May 2026].

Bourne, N.K., 2022. On the critical threshold for detonation. Combustion and Flame, 244, p.112239.

Coherent Market Insights. (2026). Industrial Explosives Market Size and Forecast, 2026–2033. Available at: https://www.coherentmarketinsights.com/market-insight/industrialexplosivesmarket-2966? (Accessed: 18 May 2026)

Djerdjev, A.M., Priyananda, P., Gore, J., Beattie, J.K., Neto, C. and Hawkett, B.S., 2021. Safer emulsion explosives resulting from NOx inhibition. Chemical Engineering Journal, 403, p.125713.

Mine Health and Safety Council (MHSC) (2021) Prevention of flammable gas explosions in mines other than coal mines. Johannesburg: MHSC. Available at:https://www.mhsc.org.za/wpcontent/uploads/2021/12/Book-1_Prevention-offlammable-gas-explosions-in-mines-otherthan-coal-m....pdf [Accessed: 10 May 2026].

MINN2012A study guide. University of Witswatersrand Micromine. (2025). Burden and Spacing. Available at: https://webhelp.micromine.com/mm/25.0/English/Content/mmblast/IDH_BLAST_BURDEN_SPACING.htm? (Accessed: 18 May 2026)

Mpofu, M. (2022). Best Practice Guidelines for the Management of Hot Holes in Opencast Coal Mines. Auckland Park: Coaltech Research Association. Available at: https://coaltech.co.za/wpcontent/uploads/2022/07/M2020-1-Management-of-hotholeblasting-Best-PracticeGuideline-for-the-Management-of-Hot-Holes-in-OpencastCoalMines.pdf [Accessed: 11 May 2026].

Mpofu, M., Maphalala, B., Kgarume, T., Magweregwede, F. and Stenzel, G. (2024) 'Development of Best Practice Guideline for the management of hot holes in surface coal mines', Journal of the Southern African Institute of Mining and Metallurgy, 124(8), pp. 473–480. Available at:https://www.saimm.co.za/Journal/v124n8p473.pdf [Accessed: 10 May 2026].

Mpofu, M., Maphalala, B., Kgarume, T., Magweregwede, F., and Stenzel, G., 2024. Development of Best Practice Guideline for the management of hot holes in surface coal mines. Journal of the Southern African Institute of Mining and Metallurgy, 124(8), pp.473482.

Oates, T.E. & Spiteri, W., 2021. 'Stemming and best practice in the mining industry: A literature review', Journal of the Southern African Institute of Mining and Metallurgy, vol. 121, no. 8, pp. 415–426.

Pearton, S.P., 2015. The application of pumpable emulsions in narrow-reef stoping. Journal of the Southern African Institute of Mining and Metallurgy, 115(6), pp.489-497.